4.8 ⋆⋆ We'll measure the puck's position by the angle θ it subtends at the sphere's center O (measured down from the top). The puck's PE (defined as zero at the level of O) is $U(\theta) = mgR\cos\theta$, and its total energy is $E = U(0) = mgR$. By conservation of energy,

$$T = \tfrac{1}{2}mv^2 = E - U = mgR(1 - \cos\theta). \qquad \text{(i)}$$

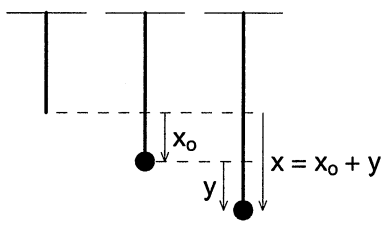
As long as the puck remains in contact with the sphere, the radial component of Newton's second law reads $N - mg\cos\theta = -mv^2/R$, where N denotes the normal force of the sphere on the puck. Substituting from Eq (i) for mv^2 we find

$$N = mg(3\cos\theta - 2).$$

As long as N is positive the puck remains on the sphere. Since the sphere cannot exert a negative normal force, once the predicted value of N becomes negative, the puck must have left the sphere. Therefore it leaves the sphere when $N = 0$ or $\theta = \arccos(2/3) = 48.2°$.

4.9 ⋆⋆ **(a)** $U = -\int_0^x F(x')dx' = k\int_0^x x'dx' = \tfrac{1}{2}kx^2$

(b) The two forces on m are gravity, mg down, and the spring, kx up, and at the new equilibrium x_o they must balance, so $x_o = mg/k$. The total PE is



$$\begin{aligned} U &= \tfrac{1}{2}kx^2 - mgx \\ &= \tfrac{1}{2}k(x_o + y)^2 - mg(x_o + y) \\ &= \tfrac{1}{2}ky^2 + y(kx_o - mg) + \text{const.} \end{aligned}$$

In the last line the second term is zero (the condition for equilibrium) and we can drop the constant, to give $U = \tfrac{1}{2}ky^2$.

4.10 ⋆

function, f	$\partial f/\partial x$	$\partial f/\partial y$	$\partial f/\partial z$
$ax^2 + bxy + cy^2$	$2ax + by$	$bx + 2cy$	0
$\sin(axyz^2)$	$ayz^2\cos(axyz^2)$	$axz^2\cos(axyz^2)$	$2axyz\cos(axyz^2)$
ae^{xy/z^2}	$(ay/z^2)e^{xy/z^2}$	$(ax/z^2)e^{xy/z^2}$	$(-2axy/z^3)e^{xy/z^2}$

4.11 ⋆

function, f	$\partial f/\partial x$	$\partial f/\partial y$	$\partial f/\partial z$
$ay^2 + 2byz + cz^2$	0	$2ay + 2bz$	$2by + 2cz$
$\cos(axy^2z^3)$	$-ay^2z^3\sin(axy^2z^3)$	$-2axyz^3\sin(axy^2z^3)$	$-3axy^2z^2\sin(axy^2z^3)$
$ar = a\sqrt{x^2 + y^2 + z^2}$	ax/r	ay/r	az/r

4.12 ⋆ **(a)** $\nabla f = \hat{\mathbf{x}}\,2x + \hat{\mathbf{z}}\,3z^2$. **(b)** $\nabla f = k\,\hat{\mathbf{y}}$. **(c)** $\nabla f = \hat{\mathbf{r}}$.

4.13 ⋆

function, f	$\partial f/\partial x$	$\partial f/\partial y$	$\partial f/\partial z$	∇f
$\ln(r)$	x/r^2	y/r^2	z/r^2	$\hat{\mathbf{r}}/r$
r^n	nxr^{n-2}	nyr^{n-2}	nzr^{n-2}	$n\,r^{n-1}\,\hat{\mathbf{r}}$
$g(r)$	$g'(r)x/r$	$g'(r)y/r$	$g'(r)z/r$	$g'(r)\,\hat{\mathbf{r}}$

4.14 ⋆ Consider the x component

$$[\nabla(fg)]_x = \frac{\partial}{\partial x}(fg) = f\frac{\partial g}{\partial x} + g\frac{\partial f}{\partial x} = f(\nabla g)_x + g(\nabla f)_x\,.$$

Since the other two components work the same way, we conclude that $\nabla(fg) = f\nabla g + g\nabla f$.

4.15 ⋆ If $f(\mathbf{r}) = x^2 + 2y^2 + 3z^2$, the gradient is $\nabla f = (2x, 4y, 6z)$, so Equation (4.35) gives the approximation

$$\begin{aligned} df \approx \nabla f\cdot d\mathbf{r} &= (2x, 4y, 6z)\cdot(dx, dy, dz) \\ &= (2, 4, 6)\cdot(0.01, 0.03, 0.05) = 0.02 + 0.12 + 0.30 = 0.4400. \end{aligned}$$

This compares favorably with the exact result

$$df = f(1.01, 1.03, 1.05) - f(1, 1, 1) = 6.4494 - 6.0000 = 0.4494.$$

4.16 ⋆ If $U = k(x^2 + y^2 + z^2)$, then $\mathbf{F} = -\nabla U = -2k(x, y, z) = -2k\mathbf{r}$.

4.17 ⋆ **(a)** The force $\mathbf{F} = q\mathbf{E}_o$ certainly doesn't depend on anything but $\mathbf{r}$ (and doesn't even depend on $\mathbf{r}$ since $\mathbf{E}_o$ is constant). The work integral is $W(1 \to 2) = \int_1^2 \mathbf{F}\cdot d\mathbf{r} = q\mathbf{E}_o\cdot\int_1^2 d\mathbf{r} = q\mathbf{E}_o\cdot(\mathbf{r}_2 - \mathbf{r}_1)$, which is independent of path. Therefore, the force is conservative, and the PE is $U(\mathbf{r}) = -W(0 \to \mathbf{r}) = -q\mathbf{E}_o\cdot\mathbf{r}$.

(b) Since $U = -q(E_{ox}x + E_{oy}y + E_{oz}z)$, $-\nabla U = q(E_{ox}, E_{oy}, E_{oz}z) = q\mathbf{E}_o = \mathbf{F}$.

4.18 ⋆⋆ **(a)** According to (4.35), the change in $f(\mathbf{r})$ resulting from any small displacement $d\mathbf{r}$ is $df = \nabla f\cdot d\mathbf{r}$. If, in particular, we consider any infinitesimal displacement $d\mathbf{r}$ in a surface of constant f, then df will be zero. This implies that $\nabla f\cdot d\mathbf{r} = \mathbf{0}$, that is, ∇f is perpendicular to the surface of constant f.

(b) Consider a displacement $d\mathbf{r} = \epsilon\mathbf{u}$ with fixed magnitude ϵ but variable direction $\mathbf{u}$. Our job is to find the direction of $\mathbf{u}$ for which the corresponding change df is largest. Since $df = \nabla f\cdot d\mathbf{r} = \epsilon\nabla f\cdot\mathbf{u} = \epsilon|\nabla f|\cos\theta$, where θ is the angle between ∇f and $\mathbf{u}$, we see that df is maximum if $\theta = 0$, or ∇f and $\mathbf{u}$ are parallel. That is, the direction of ∇f is the direction in which f increases most rapidly.

4.19 ⋆⋆ **(a)** The equation $x^2 + 4y^2 = K$ defines an ellipse in the xy plane, centered on $x = y = 0$. Since the value of $x^2 + 4y^2$ is unchanged by any variation of z, the surface is an elliptical cylinder, centered on the z axis, with semi-major and semi-minor axes equal to $\sqrt{K}$ and $\sqrt{K}/2$.

(b) We know that the vector ∇f is perpendicular to the surface $f = \text{const}$. In the present case $\nabla f = (2x, 8y, 0) = (2, 8, 0)$ at the point $(1, 1, 1)$. Therefore the unit normal is $\mathbf{n} = (1, 4, 0)/\sqrt{17}$ or $-\mathbf{n}$. The direction of maximum increase is $\mathbf{n}$.

4.20 ⋆

$\mathbf{F} =$	(kx, ky, kz)	(Ax, By^2, Cz^3)	(Ay^2, Bx, Cz)
$\nabla \times \mathbf{F} =$	$(0, 0, 0)$	$(0, 0, 0)$	$(0, 0, B - 2Ay)$

4.21 ⋆ This is exactly the same as Example 4.5 except that $\gamma = kqQ$ is replaced by $\gamma = -GMm$. Thus, if we choose $U = 0$ at $r = \infty$, then $U(r) = \gamma/r = -GMm/r$.

4.22 ⋆ Since $\mathbf{F} = \gamma\hat{\mathbf{r}}/r^2$, the spherical polar components of $\mathbf{F}$ are $F_\theta = F_\phi = 0$ and $F_r = \gamma/r^2$. Since F_r is independent of θ and ϕ, its two derivatives $\partial F_r/\partial\theta$ and $\partial F_r/\partial\phi$ are both zero. Inspection of the expression for $\nabla \times \mathbf{F}$ in spherical polars should convince you that each of the six terms is zero, so $\nabla \times \mathbf{F} = 0$. Since $\mathbf{F}$ certainly depends on $\mathbf{r}$ only, $\mathbf{F}$ is conservative.

4.23 ⋆⋆ All three given forces depend only on $\mathbf{r}$; that is, all satisfy the first condition. It remains to checks their curls.

(a) With $\mathbf{F} = k(x, 2y, 3z)$, we find $\nabla \times \mathbf{F} = (0, 0, 0)$, so $\mathbf{F}$ is conservative. The corresponding PE is $U = -\int \mathbf{F}\cdot d\mathbf{r}' = -k\int(x'dx' + 2y'dy' + 3z'dz') = -k(\frac{1}{2}x^2 + y^2 + \frac{3}{2}z^2)$. Clearly $-\nabla U = k(x, 2y, 3z) = \mathbf{F}$.

(b) With $\mathbf{F} = k(y, x, 0)$, we find $\nabla \times \mathbf{F} = (0, 0, 1-1) = (0, 0, 0)$, so $\mathbf{F}$ is conservative. The corresponding PE is $U = -\int_0^{\mathbf{r}} \mathbf{F}\cdot d\mathbf{r}' = -k\int_0^{\mathbf{r}}(y'dx' + x'dy')$. This integral can be evaluated along any path. One simple choice is to go from the origin out to x on the x axis (this contributes 0 to the integral) and then parallel to the y axis to y (which contributes $-kxy$). Thus $U = -kxy$. Clearly $-\nabla U = k(y, x, 0) = \mathbf{F}$.

(c) With $\mathbf{F} = k(-y, x, 0)$, we find $\nabla \times \mathbf{F} = (0, 0, 1+1) = (0, 0, 2)$, so $\mathbf{F}$ is not conservative.

4.24 ⋆⋆⋆ **(a)** Consider first the force on m due to a short segment dz of the rod at a height z above m. This force has magnitude $dF = Gm\mu\, dz/r^2$ in the direction shown in the left picture, where r is the distance from the element dz to m. To find the total force we must integrate this from $z = -\infty$ to ∞. When we do this, the z components F_z from points z and $-z$ will cancel. Since the component into the page is clearly zero, we have only to worry

$$\oint_\Gamma \mathbf{F}\cdot d\mathbf{r} = \int \left(\frac{\partial F_x}{\partial y} - \frac{\partial F_y}{\partial x}\right) dA = \int (\boldsymbol{\nabla}\times\mathbf{F})\cdot\hat{\mathbf{n}}\, dA.$$

4.26 ⋆ If $U = mgy$, then, whether or not g depends on t, $-\boldsymbol{\nabla}U = -mg\boldsymbol{\nabla}y = -mg(0,1,0) = \mathbf{F}$. But

$$\frac{dE}{dt} = \frac{d}{dt}\left(\tfrac{1}{2}m\mathbf{v}^2 + mgy\right) = m\dot{\mathbf{v}}\cdot\mathbf{v} + (mg\dot{y} + my\dot{g}) = -mgv_y + mgv_y + my\dot{g} = my\dot{g} \neq 0.$$

4.27 ⋆⋆ Stokes's theorem guarantees that the integral (4.48) is path-independent and hence that it defines a unique function $U(\mathbf{r},t)$. To show that $-\boldsymbol{\nabla}U = \mathbf{F}$, consider the difference between the values of U at two neighboring points $\mathbf{r}$ and $\mathbf{r}+d\mathbf{r}$:

$$dU = U(\mathbf{r}+d\mathbf{r},t) - U(\mathbf{r},t) \tag{iv}$$

$$= -\int_{\mathbf{r}}^{\mathbf{r}+d\mathbf{r}} \mathbf{F}(\mathbf{r}',t)\cdot d\mathbf{r}' = -\mathbf{F}(\mathbf{r},t)\cdot d\mathbf{r}. \tag{v}$$

But $dU = (\boldsymbol{\nabla}U)\cdot d\mathbf{r}$, and since both of these expressions for dU are valid for all (small) $d\mathbf{r}$, it follows that $\mathbf{F}(\mathbf{r},t) = -\boldsymbol{\nabla}U(\mathbf{r},t)$. The argument leading to Eq.(4.19) requires that we look at the change in $E = T + U$ as we follow the particle moving from $\mathbf{r}$ to $\mathbf{r}+d\mathbf{r}$ as the time advances from t to $t+dt$. The change in T is $dT = \mathbf{F}\cdot d\mathbf{r}$. The change in U is

$$dU = U(\mathbf{r}+d\mathbf{r},t+dt) - U(\mathbf{r},t) \tag{vi}$$

and here's the difficulty. This dU is not the same as the dU in Eq.(iv). Here we are concerned with the change dU as the particle moves from $\mathbf{r}$ to $\mathbf{r}+d\mathbf{r}$ *and* the time changes from t to $t+dt$. In Eq.(iv) dU is the difference between the values at $\mathbf{r}$ and $\mathbf{r}+d\mathbf{r}$ *at the same time* t. Thus the difference (vi) is not the same as $-\mathbf{F}(\mathbf{r},t)\cdot d\mathbf{r}$ as in (v); it does not cancel the change in KE when we evaluate $dE = dT + dU$, and mechanical energy is not conserved.

4.28 ⋆⋆ **(a)** Since $E = T + U = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$, it follows that $\dot{x}(x) = \sqrt{2/m}\sqrt{E - \frac{1}{2}kx^2}$.

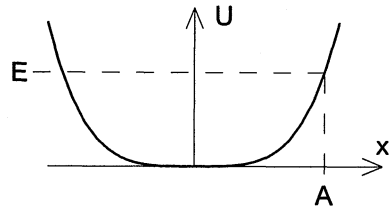
(b) At the end point $x = A$, we know that $T = 0$, so $E = \frac{1}{2}kA^2$. Substituting into the result of part (a), we find $\dot{x}(x) = \omega\sqrt{A^2 - x^2}$, where I have defined $\omega = \sqrt{k/m}$. From (4.58) we find

$$t = \int_0^x \frac{dx'}{\dot{x}(x')} = \frac{1}{\omega}\int_0^x \frac{dx'}{\sqrt{A^2 - x'^2}}.$$

The integral can be evaluated with the substitution $x' = A\sin\theta$ and gives $\arcsin(x/A)$. So $t = (1/\omega)\arcsin(x/A)$.

(c) Solving for x we find $x(t) = A\sin\omega t$. This shows that x is a sinusoidal function of t, which is the definition of simple harmonic motion. In particular, $x(t)$ repeats itself after a time t such that $\omega t = 2\pi$, or $t = 2\pi/\omega = 2\pi\sqrt{m/k}$.

4.29 ⋆⋆ **(a)** The mass moves to the right to the turning point at $x = A$, where $E = U = kA^4$ so that T and hence v are zero. It speeds up as it moves back to O and then moves out to the left until $x = -A$, where $E = U$ again. It then moves back to O, and repeats the whole cycle indefinitely (in the absence of any friction). Notice how this quartic well is much flatter at the bottom than the parabolic harmonic well.



(b) Since $T = E - U = k(A^4 - x^4)$, the velocity is $\dot{x} = \sqrt{2k/m}\sqrt{A^4 - x^4}$ (moving to the right). According to (4.58), the time to move out to $x = A$ is

$$t(0 \to A) = \int_0^A \frac{dx}{\dot{x}} = \sqrt{\frac{m}{2k}} \int_0^A \frac{dx}{\sqrt{A^4 - x^4}}.$$

(c) The period is four times this time. Thus, if we change variables to $u = x/A$,

$$\tau = \frac{1}{A}\sqrt{\frac{8m}{k}} \int_0^1 \frac{du}{\sqrt{1 - u^4}}$$

which is inversely proportional to A as claimed.

(d) Using any suitable software, the integral can be found to be 1.31, and setting $m = k = A = 1$ we find $\tau = 3.71$.

4.30 ⋆ **(a)** As the toy tips, the hemisphere rolls and its center O remains at a fixed height. On the other hand the height of the CM above O changes from $h - R$ to $(h - R)\cos\theta$. Therefore, the PE of the toy is now $U(\theta) = mg[R + (h - R)\cos\theta]$.

(b) Since $dU/d\theta = -mg(h - R)\sin\theta$, which vanishes at $\theta = 0$, we see that the upright position is an equilibrium, as expected. Next, $d^2U/d\theta^2 = -mg(h - R)\cos\theta = mg(R - h)$ at $\theta = 0$. Thus the equilibrium is stable if and only if $R > h$. [If $R = h$, then $U(\theta) = mgR =$ const, and the equilibrium is neutral.]

4.31 ⋆ **(a)** Because the string is inextensible, the height of m_2 below the wheel is $y = k - x$, where k is a constant, and its velocity is $\dot{y} = -\dot{x}$. Thus the total energy is $E = T + U = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{x}^2 - m_1gx - m_2g(k - x)$ or $E = \frac{1}{2}(m_1 + m_2)\dot{x}^2 - (m_1 - m_2)gx$, if we drop the uninteresting constant $-m_2gk$.

(b) The equation $dE/dt = 0$ yields $(m_1 + m_2)\dot{x}\ddot{x} - (m_1 - m_2)g\dot{x} = 0$, or

$$(m_1 + m_2)\ddot{x} = (m_1 - m_2)g. \tag{vii}$$

Applying the second law to each separate mass, we find

$$m_1\ddot{x} = m_1g - F_T \quad \text{and} \quad m_2\ddot{x} = F_T - m_2g$$

where F_T is the tension in the string. Adding these two equations, we can eliminate the tension and we get precisely Eq.(vii).

4.32 ⋆⋆ **(a)** Consider an infinitesimal displacement $d\mathbf{r} = (dx, dy, dz)$ occurring in a time dt. The bead's velocity is $\mathbf{v} = (dx/dt, dy/dt, dz/dt)$, so the speed is

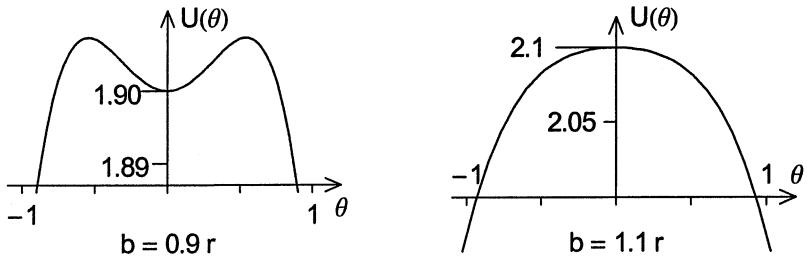
$$v = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = \frac{\sqrt{dx^2 + dy^2 + dz^2}}{dt} = \frac{ds}{dt} = \dot{s}$$

(b) Differentiating the equation $v^2 = \mathbf{v}^2$, we get $2v\dot{v} = 2\mathbf{v}\cdot\dot{\mathbf{v}} = 2v\hat{\mathbf{v}}\cdot\mathbf{F}/m$. Cancelling the factor of $2v$ and recognising from part (a) that $\dot{v} = \ddot{s}$, we find that $m\ddot{s} = \hat{\mathbf{v}}\cdot\mathbf{F} = F_{\text{tang}}$

(c) The net force on the particle is $\mathbf{F} = \mathbf{N} - \boldsymbol{\nabla}U$. Therefore, $F_{\text{tang}} = -\hat{\mathbf{v}}\cdot\boldsymbol{\nabla}U$ (since $\mathbf{N}$ is perpendicular to $\hat{\mathbf{v}}$). Now, if we imagine a small displacement ds along the wire, $d\mathbf{r} = ds\,\hat{\mathbf{v}}$ (since $\hat{\mathbf{v}}$ is a unit vector tangent to the wire), so $dU = d\mathbf{r}\cdot\boldsymbol{\nabla}U = ds\,\hat{\mathbf{v}}\cdot\boldsymbol{\nabla}U$. Therefore, $F_{\text{tang}} = -\hat{\mathbf{v}}\cdot\boldsymbol{\nabla}U = -dU/ds$.

4.33 ⋆⋆ **(a)** The derivation of $U(\theta)$ is given above Eq.(4.59).

(b)



(c) These two plots do bear out the finding of Example 4.7 that for $b < r$ the equilibrium at $\theta = 0$ is stable, whereas for $b > r$ it is unstable. In addition, they show that for $b < r$ there can be two further equilibrium points (symmetrically placed on either side of $\theta = 0$), both of which are unstable

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4.34 ⋆⋆ **(a)** The distance of the mass m below the support is $l\cos\phi$. Therefore, its height measured up from the equilibrium position is $l - l\cos\phi = l(1 - \cos\phi)$ and its PE is $U = mgl(1 - \cos\phi)$. The total energy is $E = \frac{1}{2}ml^2\dot{\phi}^2 + mgl(1 - \cos\phi)$.

(b) The equation $dE/dt = 0$ reads $ml^2\dot{\phi}\ddot{\phi} + mgl\dot{\phi}\sin\phi = 0$ or $ml^2\ddot{\phi} = -mgl\sin\phi$. That is, $I\alpha = \Gamma$.

(c) Provided ϕ remains small, the equation of motion is well-approximated by $l\ddot{\phi} = -g\phi$, whose solution is $\phi = A\cos(\omega t) + B\sin(\omega t)$, where $\omega = \sqrt{g/l}$. This has period $\tau_o = 2\pi\sqrt{l/g}$.

4.35 ⋆⋆ **(a)** Because the string is inextensible, the height of m_2 below the wheel is $y = k - x$, where k is a constant, and its velocity is $\dot{y} = -\dot{x}$. The angular velocity of the wheel is $\omega = \dot{x}/R$. Thus the total energy is

$$\begin{aligned} E = T + U &= \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\dot{x}^2 + \tfrac{1}{2}I\omega^2 - m_1gx - m_2g(k - x) \\ &= \tfrac{1}{2}(m_1 + m_2 + I/R^2)\dot{x}^2 - (m_1 - m_2)gx, \end{aligned}$$

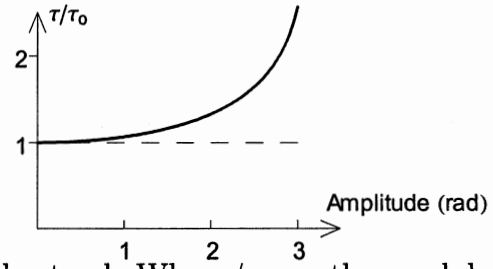
(To get the last expression I used $\sqrt{l/g} = \tau_o/2\pi$.) The time to swing from $\phi = 0$ to Φ is given by the integral $\int d\phi/\dot{\phi}$ taken from 0 to Φ (using the positive square root for $\dot{\phi}$). Since the period is four times this amount, we conclude that

$$\tau = 4\int_0^\Phi \frac{d\phi}{\dot{\phi}} = \frac{\tau_o}{\pi}\int_0^\Phi \frac{d\phi}{\sqrt{\sin^2\Phi/2 - \sin^2\phi/2}}.$$

If we make the recommended substitutions, $\sin\Phi/2 = A$ and $\sin\phi/2 = Au$, the square root in the denominator becomes $A\sqrt{1-u^2}$ and $d\phi = 2A\,du/\cos\phi/2 = 2A\,du/\sqrt{1-A^2u^2}$. Putting all this together, we find

$$\tau = \tau_o\frac{2}{\pi}\int_0^1 \frac{du}{\sqrt{1-u^2}\sqrt{1-A^2u^2}} = \tau_o\frac{2}{\pi}K(A^2). \qquad \text{(x)}$$

(b) When the amplitude is small (less than about 0.5 rad), τ is very close to τ_o, as we would expect. When the amplitude is 3 rad (about 170 degrees), τ is about 2.5 times longer than its small-amplitude value. The graph suggests that as the amplitude approaches π, the period may approach infinity, a conclusion that is confirmed by putting $A = 1$ in the integral (x). This is easy to understand: When $\phi = \pi$, the pendulum is actually in unstable equilibrium. Thus the closer it gets to π, the more slowly it moves, causing the period to approach infinity.



4.39 ⋆⋆⋆ **(a)** Same as 4.38(a). **(b)** Ignoring the second square root completely, we get

$$\tau = \tau_o\frac{2}{\pi}\int_0^1 \frac{du}{\sqrt{1-u^2}} = \tau_o$$

as expected. (To show the integral is just $\pi/2$, use the substitution $u = \sin\alpha$.)

(c) If we make the binomial approximation suggested, we get

$$\tau = \tau_o\frac{2}{\pi}\left(\int_0^1 \frac{du}{\sqrt{1-u^2}} + \tfrac{1}{2}A^2\int_0^1 \frac{u^2\,du}{\sqrt{1-u^2}}\right).$$

The first integral is $\pi/2$ as before, while the same substitution shows the second integral to be $\pi/4$. Recalling that $A = \sin\Phi/2$, we find that $\tau = \tau_o[1 + \frac{1}{4}\sin^2(\Phi/2)]$, as stated. If $\Phi = 45°$, this gives $\tau = 1.037\tau_o$, which represents a 3.7% correction to the small-amplitude approximation (τ_o), and is itself within 0.3% of the exact answer ($1.040\tau_o$).

4.40 ⋆ **(a)** Referring to Fig.4.16, notice that the side OQ has length $OQ = r\sin\theta$, thus

$$x = OQ\cos\phi = r\sin\theta\cos\phi, \quad y = OQ\sin\phi = r\sin\theta\sin\phi, \quad z = r\cos\theta.$$

(b) By Pythagoras' theorem, $r = \sqrt{x^2 + y^2 + z^2}$. By elementary trig,

$$\theta = \arccos(z/r) = \arccos(z/\sqrt{x^2 + y^2 + z^2}) \text{ and } \phi = \arctan(y/x).$$

(But note that this leaves an ambiguity of π in ϕ. See the solution to Problem 1.42.)

4.41 ⋆ Since the mass moves in a circle, the radial component of its accelerations is just $-v^2/r$, the centripetal acceleration. By Newton's second law, $m(-v^2/r) = F_r = -dU/dr = -nkr^{n-1}$, from which we see that $mv^2 = nkr^n = nU$, and hence $T = \frac{1}{2}mv^2 = \frac{1}{2}nU$.

4.42 ⋆ Assuming that the spring doesn't bend, its extension is $r - r_o$ and the force which it exerts is $\mathbf{F} = -k(r - r_o)\hat{\mathbf{r}}$. Because this is central and spherically symmetric, it is automatically conservative.

4.43 ⋆⋆ **(a)** Since $\hat{\mathbf{r}} = \mathbf{r}/r$ we can write $\mathbf{F} = f(r)\mathbf{r}/r = g(r)\mathbf{r}$, say, where $g(r)$ is simply a convenient new name for $f(r)/r$. Now consider the x component of $\nabla \times \mathbf{F}$:

$$\begin{aligned}(\nabla \times \mathbf{F})_x &= \partial_y F_z - \partial_z F_y \\ &= \partial_y[g(r)z] - \partial_z[g(r)y] = z\,\partial_y g(r) - y\,\partial_z g(r).\end{aligned}$$

Now by the chain rule, $\partial_y g(r) = g'(r)\partial r/\partial y = g'(r)y/r$, and similarly for $\partial_z g(r)$. Therefore

$$(\nabla \times \mathbf{F})_x = zg'(r)y/r - yg'(r)z/r = 0.$$

Since the other two components work the same way, we conclude that $\nabla \times \mathbf{F} = 0$ and $\mathbf{F}(\mathbf{r})$ is conservative.

(b) If $\mathbf{F}(\mathbf{r})$ is central and spherically symmetric, its spherical polar components are $F_r = f(r)$ and $F_\theta = F_\phi = 0$. When we insert these in the given expression for $\nabla \times \mathbf{F}$ in spherical polars, only the two terms involving F_r survive since $F_\theta = F_\phi = 0$, but both of these remaining terms involve derivatives of F_r with respect to θ or ϕ, which are also zero. Therefore $\nabla \times \mathbf{F} = 0$.

4.44 ⋆⋆ Since $\mathbf{F} = f(r)\hat{\mathbf{r}}$, the work done going radially out from A to C is $W_{AC} = \int_A^C \mathbf{F}\cdot d\mathbf{r} = \int_{r_A}^{r_B} f(r)dr$. The same argument applies to W_{DB}, so $W_{AC} = W_{DB}$. On the other hand, on the paths CB and AD, $\mathbf{F}$ is perpendicular to $d\mathbf{r}$, so $W_{CB} = W_{AD} = 0$. Therefore

$$W_{ACB} = W_{AC} + W_{CB} = W_{AD} + W_{DB} = W_{ADB}$$

4.45 ⋆⋆ That $\mathbf{F}$ is conservative tells us that the amounts of work done along the two paths ACB and ADB are equal, $W_{ACB} = W_{ADB}$. That $\mathbf{F}$ is central implies that no work is done along CB and AD, that is, $W_{CB} = W_{AD} = 0$. Therefore $W_{AC} = W_{DB}$. Since $\mathbf{F} = f(\mathbf{r})\hat{\mathbf{r}}$, $W_{AC} = f(\mathbf{r}_A)dr$ and $W_{DB} = f(\mathbf{r}_D)dr$, and since these two are equal, it follows that $f(\mathbf{r}_A) = f(\mathbf{r}_D)$. Finally, since A and D are any two points at the same distance from O, this says that $f(\mathbf{r})$ depends only on $|\mathbf{r}|$. That is, $f(\mathbf{r}) = f(r)$.

4.46 ⋆ In an elastic collision KE is conserved, so

$$m_1 v_1^2 = m_1 v_1'^2 + m_2 v_2'^2. \tag{xi}$$

Since momentum is also conserved, $m_1\mathbf{v}_1 = m_1\mathbf{v}_1' + m_2\mathbf{v}_2'$, or, squaring both sides,

$$m_1^2 v_1^2 = m_1^2 v_1'^2 + m_2^2 v_2'^2 + 2m_1 m_2 \mathbf{v}_1' \cdot \mathbf{v}_2'. \tag{xii}$$

If we multiply both sides of (xi) by m_1 and subtract from (xii), four terms cancel and we're left with

$$0 = m_2(m_2 - m_1)v_2'^2 + 2m_1 m_2 \mathbf{v}_1' \cdot \mathbf{v}_2'.$$

Therefore

$$\mathbf{v}_1' \cdot \mathbf{v}_2' = \frac{m_1 - m_2}{2m_1} v_2'^2.$$

If $m_1 > m_2$, $\cos\theta$ has to be positive, so $\theta < \pi/2$. Whereas if $m_1 < m_2$, $\cos\theta$ has to be negative, so $\theta > \pi/2$.

4.47 ⋆ Since the problem is one-dimensional, we can use v to denote velocity (strictly speaking the x component of velocity). Then conservation of KE (a little rearranged) says

$$m_1(v_1^2 - v_1'^2) = m_2(v_2'^2 - v_2^2)$$

and, similarly, conservation of momentum

$$m_1(v_1 - v_1') = m_2(v_2' - v_2).$$

Dividing the first of these by the second, we find

$$v_1 + v_1' = v_2' + v_2 \quad \text{or} \quad v_1 - v_2 = v_2' - v_1'.$$

4.48 ⋆ Let the initial speed of particle 1 be v_1 and the final speed of the composite be v'. Then, conservation of momentum says that $m_1 v_1 = (m_1 + m_2)v'$. Therefore the initial and final KEs are $T = \frac{1}{2}m_1 v_1^2$ and $T' = \frac{1}{2}(m_1 + m_2)v'^2 = \frac{1}{2}m_1^2 v_1^2/(m_1 + m_2)$, and the fractional loss of KE is

$$\frac{T - T'}{T} = \frac{m_1(m_1 + m_2) - m_1^2}{m_1(m_1 + m_2)} = \frac{m_2}{m_1 + m_2}.$$

If $m_1 \ll m_2$, almost all the initial KE is lost; if $m_2 \ll m_1$, almost none of the initial KE is lost.

4.49 ⋆⋆ Let's define $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, so that $\mathbf{r}$ is a vector pointing from particle 2 to particle 1. The PE is $U = \gamma/r$, the force on particle 1 due to particle 2 is $\mathbf{F}_{12} = (\gamma/r^2)\hat{\mathbf{r}}$, and that on particle 2 due to particle 1 is $\mathbf{F}_{21} = -(\gamma/r^2)\hat{\mathbf{r}}$. We'll start with the x component of $-\boldsymbol{\nabla}_1 U$:

$$(-\boldsymbol{\nabla}_1 U)_x = -\frac{\partial U}{\partial x_1} = \frac{\gamma}{r^2}\frac{\partial r}{\partial x_1} = \frac{\gamma}{r^3}(x_1 - x_2) \tag{xiii}$$

where for the second equality I used the chain rule and for the third I used the following:

$$\frac{\partial r}{\partial x_1} = \frac{\partial}{\partial x_1}\sqrt{(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2} = \frac{x_1-x_2}{\sqrt{\cdots}} = \frac{x_1-x_2}{r}. \qquad \text{(xiv)}$$

Combining (xiii) with the corresponding results for the y and z components, we see that

$$-\boldsymbol{\nabla}_1 U = \frac{\gamma}{r^3}(\mathbf{r}_1-\mathbf{r}_2) = \frac{\gamma}{r^3}\mathbf{r} = \frac{\gamma}{r^2}\hat{\mathbf{r}} = \mathbf{F}_{12}. \qquad \text{(xv)}$$

If we had evaluated $-\boldsymbol{\nabla}_2 U$, the only difference would have been that in the equivalent of (xiv), we would have had an extra minus sign because $\partial(x_1-x_2)^2/\partial x_2 = -2(x_1-x_2)$. Thus in place of (xv), we would have found $-\boldsymbol{\nabla}_2 U = -(\gamma/r^2)\hat{\mathbf{r}} = \mathbf{F}_{21}$.

4.50 ⋆⋆ By the chain rule

$$\frac{\partial}{\partial x_1} f(x_1-x_2) = f'(x_1-x_2)\frac{\partial}{\partial x_1}(x_1-x_2) = f'(x_1-x_2)$$

where f' denotes the derivative of f with respect to its argument. On the other hand,

$$\frac{\partial}{\partial x_2} f(x_1-x_2) = f'(x_1-x_2)\frac{\partial}{\partial x_2}(x_1-x_2) = -f'(x_1-x_2).$$

This establishes the required result for the one-dimensional function $f(x_1-x_2)$. We can prove the corresponding three-dimensional result, one component at a time. For example, $(\boldsymbol{\nabla}_1 U)_x = \partial U/\partial x_1$, whereas $(\boldsymbol{\nabla}_2 U)_x = \partial U/\partial x_2$, and by the one-dimensional result, the latter is minus the former. (The four extra variables y_1, z_1, y_2, z_2 act as constants and do not affect the result.) Since the y and z components work the same way, this establishes the required three-dimensional result.

4.51 ⋆⋆

$$\begin{aligned} U(\mathbf{r}_1,\mathbf{r}_2,\mathbf{r}_3,\mathbf{r}_4) = {} & U_{12}(\mathbf{r}_1-\mathbf{r}_2)+U_{13}(\mathbf{r}_1-\mathbf{r}_3)+U_{14}(\mathbf{r}_1-\mathbf{r}_4) \\ & +U_{23}(\mathbf{r}_2-\mathbf{r}_3)+U_{24}(\mathbf{r}_2-\mathbf{r}_4)+U_{34}(\mathbf{r}_3-\mathbf{r}_4) \\ & +U_1^{\text{ext}}(\mathbf{r}_1)+U_2^{\text{ext}}(\mathbf{r}_2)+U_3^{\text{ext}}(\mathbf{r}_3)+U_4^{\text{ext}}(\mathbf{r}_4) \end{aligned}$$

so

$$\begin{aligned} -\boldsymbol{\nabla}_3 U = {} & 0+\mathbf{F}_{31}+0 \\ & +\mathbf{F}_{32}+0+\mathbf{F}_{34} \\ & +0+0+\mathbf{F}_3^{\text{ext}}+0 = \mathbf{F}_3^{\text{net}}. \end{aligned}$$

4.52 ⋆⋆ **(a)** By the work-KE theorem for each of the four particles, $dT_1 = W_1, dT_2 = W_2, dT_3 = W_3$, and $dT_4 = W_4$. Adding these four equations, we conclude that $dT = W_{\text{tot}}$.

(b) The work done on the four separate particles has the form

$$W_1 = (\mathbf{F}_{12} + \mathbf{F}_{13} + \mathbf{F}_{14} + \mathbf{F}_1^{\text{ext}})\cdot d\mathbf{r}_1$$
$$W_2 = (\mathbf{F}_{21} + \mathbf{F}_{23} + \mathbf{F}_{24} + \mathbf{F}_2^{\text{ext}})\cdot d\mathbf{r}_2$$
$$W_3 = (\mathbf{F}_{31} + \mathbf{F}_{32} + \mathbf{F}_{34} + \mathbf{F}_3^{\text{ext}})\cdot d\mathbf{r}_3$$
$$W_4 = (\mathbf{F}_{41} + \mathbf{F}_{42} + \mathbf{F}_{43} + \mathbf{F}_4^{\text{ext}})\cdot d\mathbf{r}_4$$

When we add these, we get 12 terms involving internal forces which we can group in pairs ($\mathbf{F}_{12}$ with $\mathbf{F}_{21} = -\mathbf{F}_{12}$ and so on), to give

$$\begin{aligned} W = {} & \mathbf{F}_{12}\cdot(d\mathbf{r}_1 - d\mathbf{r}_2) + \mathbf{F}_{13}\cdot(d\mathbf{r}_1 - d\mathbf{r}_3) + \mathbf{F}_{14}\cdot(d\mathbf{r}_1 - d\mathbf{r}_4) \\ & + \mathbf{F}_{23}\cdot(d\mathbf{r}_2 - d\mathbf{r}_3) + \mathbf{F}_{24}\cdot(d\mathbf{r}_2 - d\mathbf{r}_4) + \mathbf{F}_{34}\cdot(d\mathbf{r}_3 - d\mathbf{r}_4) \\ & + \mathbf{F}_1^{\text{ext}}\cdot d\mathbf{r}_1 + \mathbf{F}_2^{\text{ext}}\cdot d\mathbf{r}_2 + \mathbf{F}_3^{\text{ext}}\cdot d\mathbf{r}_3 + \mathbf{F}_4^{\text{ext}}\cdot d\mathbf{r}_4. \end{aligned}$$

The first term on the right is equal to $-dU_{12}$ [exactly as in Equations (4.85) and (4.86)], and the next five terms behave similarly. The first term on the third line is just $-dU_1^{\text{ext}}$, and so on. Thus,

$$W = (-dU_{12} - dU_{13} - \cdots - dU_{34}) + (-dU_1^{\text{ext}} - \cdots - dU_4^{\text{ext}}) = -dU,$$

as required. Combining these two results, we have $dT = -dU$ or $E = T + U = \text{const}$.

4.53 ⋆⋆ **(a)** The centripetal acceleration is supplied by the Coulomb force, so $mv^2/r = ke^2/r^2$ and thence $T = \frac{1}{2}mv^2 = \frac{1}{2}ke^2/r = -\frac{1}{2}U$.

(b) $E = T_1 + T_2 + U_1 + U_2 + U_{12} = \frac{1}{2}m_1{v_1}^2 + \frac{1}{2}m_2{v_2}^2 - ke^2\left(\frac{1}{r_1} + \frac{1}{r_2} - \frac{1}{r_{12}}\right)$.

(c) Long before: $E = T_1 + T_2 + U_1 + 0 + 0 = T_2 - \dfrac{ke^2}{2r}$.

Long after: $E' = T_1' + T_2' + 0 + U_2' + 0 = T_1' - \dfrac{ke^2}{2r'}$.

Therefore, by conservation of energy, $T_1' = T_2 + \frac{1}{2}ke^2\left(\dfrac{1}{r'} - \dfrac{1}{r}\right)$.
